

BNO055 Bosch-Sensortec C Library bus_read



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AI Overview



To integrate the Bosch BNO055 C Library with an FT260 USB-to-I2C bridge, you must map Bosch's generic `bus_write` and `bus_read` function pointers to the FT260 I2C master API calls (found in `LibFT260.h`). Bosch Sensortec Community +3

Because BNO055 requires a repeated-start condition for reading registers, the `bus_read` function must first write the target register address, and then read the requested data length. YouTube · STM32World

Implementation

Include the FTDI library and define the function mappings: FTDI

C

```
#include <windows.h>
#include <stdio.h>
#include "LibFT260.h"
#include "bno055.h" // Bosch BNO055 Header

// Global FT260 Handle
FT260_HANDLE ft260_handle = NULL;
uint8_t BNO055_I2C_ADDR = 0x28; // Default BNO055 I2C Address

/*
 * bus_read function mapped to the BNO055 Library
 */
s8 bno055_bus_read(u8 dev_addr, u8 reg_addr, u8 *reg_data, u8 cnt) {
    FT260_STATUS status;
    uint16_t bytes_read = 0;

    // 1. Write the target register address (no stop bit, preparing for repeated-
    status = FT260_I2CMaster_Write(ft260_handle,
                                   dev_addr,
```

```

        FT260_I2C_START,
        &reg_addr,
        1);

if (status != FT260_OK) {
    return -1; // Error handling
}

// 2. Read the requested number of bytes from the device
status = FT260_I2CMaster_Read(ft260_handle,
                               dev_addr,
                               FT260_I2C_REPEATED_START_STOP,
                               reg_data,
                               cnt,
                               &bytes_read);

if (status != FT260_OK || bytes_read != cnt) {
    return -1;
}

return 0; // Success
}

/*

```

Ask anything



```

buffer[0] = reg_addr;
memcpy(&buffer[1], reg_data, cnt);

status = FT260_I2CMaster_Write(ft260_handle,
                                dev_addr,
                                FT260_I2C_START_AND_STOP,
                                buffer,
                                cnt + 1);

if (status != FT260_OK) {
    return -1;
}

```

```

    return 0; // Success
}

/*
 * Delay function mapped to the BNO055 Library
 */
void bno055_delay_msec(u32 msec) {
    Sleep(msec);
}

```

Use code with caution.



Library Initialization & Binding

Before calling Bosch's `bno055_init()`, make sure to open the FT260 device, reset the I2C master, and link your custom functions to the `bno055_t` structure:  Bosch Sensortec Community +1

C

```

void init_bno055_device() {
    // 1. Initialize FT260 and assign ft260_handle...
    // FT260_OpenByVidPid(...);

    // 2. Reset and set I2C clock to 100kHz
    FT260_I2CMaster_Reset(ft260_handle);
    FT260_I2CMaster_Init(ft260_handle, 100);

    // 3. Bind functions to Bosch's structure
    struct bno055_t bno055_dev;
    bno055_dev.bus_read = bno055_bus_read;
    bno055_dev.bus_write = bno055_bus_write;
    bno055_dev.delay_msec = bno055_delay_msec;
    bno055_dev.dev_addr = BNO055_I2C_ADDR;

    // 4. Initialize the BNO055 API
    bno055_init(&bno055_dev);
}

```

Use code with caution.



Useful Resources

- For detailed information on FT260 report formats, consult the [FTDI FT260 User Guide](#).
- To download the official C source code framework, refer to the [Bosch BNO055 GitHub Repository](#). 

How to use BNO055 API's - Bosch Sensortec Community

Mar 25, 2021 — But then when I initialize the API using bno055_init(), I am only able to do this once with only one device's information. Any sug...



Bosch Sensortec Community



GitHub - boschsensortec/BNO055_SensorAPI: Bosch Sensortec BNO055 sensor driver. To report issues, go to https://community.bosch-...

Jan 13, 2020 — The Bosch Sensortec MEMS BNO055 sensor driver is a package that includes the following files: * **bno055.h** This header file cont...



BOSCH



GitHub



BNO055_SensorAPI/README.md at master - GitHub

Driver files information * bno055.h. This header file has the register address definition, constant definitions, data type definit...



GitHub



Show all

AI can make mistakes, so double-check responses



Bosch Sensortec Community

<https://community.bosch-sensortec.com> › post › how-to...

How to use BNO055 API's

Mar 25, 2021 — We mainly refer to the interface transplant in the following **function**, set the device address of **BNO055**, implement the read and write interface ... [Read more](#)

BNO055 sensor library using

Sep 4, 2023

Using **BNO055** over I2C

Mar 25, 2021

[More results from community.bosch-sensortec.com](#)Missing: ~~Library~~ ft260

GitHub

[https://github.com › boschsensortec › BNO055_SensorAPI](https://github.com/boschsensortec/BNO055_SensorAPI)**GitHub - boschsensortec/BNO055_SensorAPI: Bosch ...**

This file shall be used as an user guidance, here you can find samples of Initialize the sensor with I2C communication, Add your code to the I2C **bus_read** and ... [Read more](#)

Missing: ~~ft260~~ | Show results with: **ft260**

Adafruit Forums

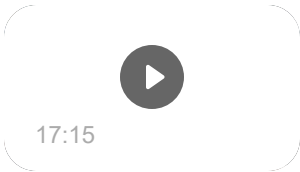
[https://forums.adafruit.com › viewtopic](https://forums.adafruit.com/viewtopic)**bn055: Using Bosch's C Library**

Oct 2, 2015 — It's a very powerful **library** and can make your life a lot easier if you want to use adafruit's **bn055** breakout board but you are not using an Arduino ... [Read more](#)

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YouTube · Dr Jon EA: Pico & Pico 2

3K+ views · 3 months ago

The BNO055: At Last, an IMU I Can Tame Quickly - C/C++

I've got this BNO055 here measuring your pitch and roll plus acceleration on the X Y and Z axis.

17:15

9 key moments in this video ▼

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GitHub

[https://github.com › BNO055_driver › blob › bno055_s...](https://github.com/BNO055_driver/blob/bno055_s...)**BNO055_SensorAPI/bno055_support.c** at master

/* Based on the user need configure I2C interface. * It is example code to explain how to use the **bn055** API*/. I2C_routine();. [Read more](#)

Missing: ~~Library~~ ft260

Reddit · r/embedded

2 comments · 2 years ago

BNO055 sensor driver with STM32CubeIDE : r/embedded

I'm trying to understand the driver shared by **Bosch Sensortec** for my IMU sensor, **BNO055** (https://github.com/boschsensortec/BNO055_driver). I ... [Read more](#)

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TI E2E support forums

<https://e2e.ti.com> › cfs-file › __key › 1738.Guide-...

How to Integrate the Bosch BNO055 ^{PDF} Library into your Tiva ...

Like, if you want to write a byte you have to send a start bit then, then the slave address, then get an ack back from the **bno055**, then send the register ... [Read more](#)

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What replaced the Bosch BNO055?



MathWorks

<https://la.mathworks.com> › ... › Sensors

Read and Plot Real-Time Data from BNO055 Sensor in ...

This example shows you how to read and plot calibrated data from a **Bosch Sensortec BNO055** a 9-axis absolute orientation sensor in real time.



ESP32 Forum

<https://esp32.com> › ... › Discussion Forum › ESP-IDF

BNO055 Bosch sensortec library integration

Apr 22, 2025 — I sought to use the original **library** provided by **Bosch**, but I am not confident my read and write **function** implementations are correct. [Read more](#)

Missing: ~~bus_read ft260~~



All About Circuits

<https://www.allaboutcircuits.com> › Projects

Capturing IMU Data with a BNO055 Absolute Orientation ...

Mar 22, 2017 — The **BNO055** is an absolute orientation sensor from **Bosch** that combines sensor data and a microprocessor to filter and combine the data. [Read more](#)

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